

NMEC595 – Research Methodology

Lecture 13: Hypothesis Testing

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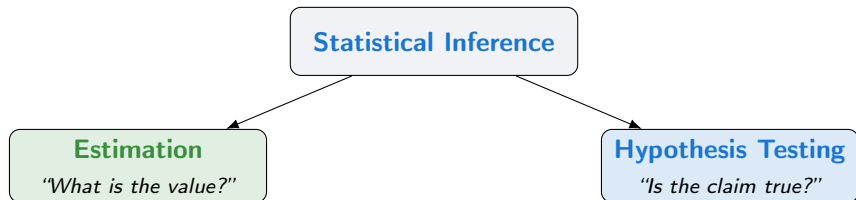
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Outline

- 1 Introduction to Hypothesis Testing
- 2 Errors, α , β and Power
- 3 The Six-Step Framework
- 4 One-Sample Proportion Test
- 5 One-Sample Mean Test
- 6 Confidence Intervals and Two-Sided Tests
- 7 Common Pitfalls & Quick Reference
- 8 Hypothesis Test for One Mean
- 9 Further Considerations
- 10 Summary

Estimation to Hypothesis Testing

Two Branches of Statistical Inference:



Estimation

- Produced a confidence interval
- Example: *"Approval rating is between 41.5% and 46.5%"*

Hypothesis Testing

- Test a specific claim
- Example: *"Is approval rating above 40%?"*

Learning Objectives

By the end of this lecture you should be able to:

- 1 **Explain** the logic and terminology of hypothesis testing
- 2 **Set up** null and alternative hypotheses correctly
- 3 **Perform** the one-proportion z -test (p-value *and* rejection-region approaches)
- 4 **Perform** the one-mean t -test
- 5 **Connect** a two-sided test to a confidence interval
- 6 **Interpret** Type I / Type II errors, α , β , and Power

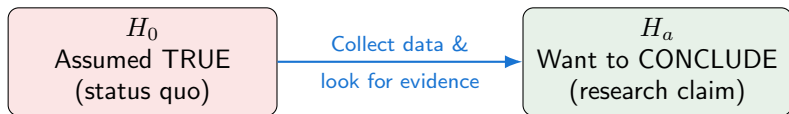
Two Competing Hypotheses

Null Hypothesis (H_0):

The “**status quo**” statement. Assumed **true** until there is strong evidence against it.

Alternative Hypothesis (H_a or H_1):

The **research / claim** we are trying to support. Also called the *research hypothesis*.



Intuition: Think of a trial: the defendant is assumed innocent (H_0) and the prosecution tries to prove guilt (H_a).

Example 1

Scenario:

Mr. OJ is on trial for murder. He is either **guilty** or **innocent**. Set up the hypotheses.

Hypotheses:

H_0 : Mr. OJ is **innocent**

H_a : Mr. OJ is guilty

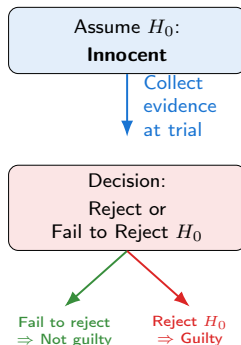
Why This Setup?

We *assume innocence* until proven guilty.

The prosecution must provide **strong evidence** to overcome the presumption of innocence.

This protects against convicting innocent people (Type I error).

Intuition: This mirrors the legal principle: “Innocent until proven guilty beyond reasonable doubt.” The “reasonable doubt” is our significance level α .



Possible Outcomes and The Golden Rule

Golden Rule:

We **NEVER** say “accept H_0 ”. We only say **reject** H_0 or **fail to reject** H_0 .

Decision	Truth (unknown)	
	H_0 is TRUE	H_0 is FALSE
Reject H_0	???	Correct ✓
Fail to reject H_0	Correct ✓	???

What happens in the “???” cells? $\Rightarrow$ We call those **errors**. See next slide.

Types of Error

Decision	H_0 is TRUE	H_0 is FALSE
Reject H_0 (True Positive)	Type I Error (False Positive)	Correct ✓
Fail to reject H_0 (False Negative)	Correct ✓ (True Negative)	Type II Error

Example 2

Recall: H_0 : Innocent H_a : Guilty

Type I Error (α):

Convict an **innocent** person

$$\alpha = P(\text{convict} \mid \text{innocent})$$

Most serious! An innocent person goes to jail.

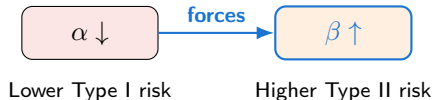
Type II Error (β):

Free a **guilty** person

$$\beta = P(\text{free} \mid \text{guilty})$$

Less serious (from a justice system perspective).

The Trade-off (for fixed n):



Fix the trade-off by **increasing** sample size n .

Try It

Problem:

An inspector certifies a building as **safe** or **not safe**. Set up hypotheses and interpret both error types.

Hypotheses: H_0 : Building is **NOT safe** vs H_a : Building is safe

Type I Error

Certify an unsafe building as safe.

Could cause deaths – extremely serious!

Type II Error

Reject a safe building.

Economic loss but no physical harm.

Why set H_0 as not safe?

We want to *assume danger* until safety is proven. This way, a Type I error (falsely certifying danger as safe) is the error we control via α .

Set up hypotheses

- We will examine two common hypothesis tests: one for a single **proportion** and one for a single **mean**. Each test involves setting up hypotheses and calculating a test statistic.
- Hypothesis tests are used to make inferences about population parameters. The alternative (research) hypothesis reflects the claim we wish to investigate.
- Hypotheses are written using parameter notation: p for proportion and μ for mean. The specific form of the alternative hypothesis depends on the research question.
- Below are the general forms of hypotheses, where p_0 and μ_0 represent the hypothesized parameter values.

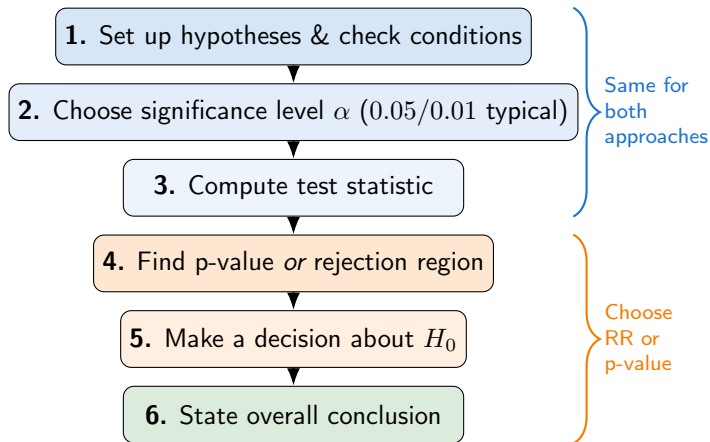
Set up hypotheses & check conditions

How to Set Up Hypotheses?

Key rule: The *alternative* H_a is what you want to **prove**. The *null* H_0 always contains an equality (=).

Research Question	H_0	H_a	Test Type
Is p different from p_0 ?	$p = p_0$	$p \neq p_0$	Two-tailed
Is p greater than p_0 ?	$p = p_0$	$p > p_0$	Right-tailed
Is p less than p_0 ?	$p = p_0$	$p < p_0$	Left-tailed
Is μ different from μ_0 ?	$\mu = \mu_0$	$\mu \neq \mu_0$	Two-tailed
Is μ greater than μ_0 ?	$\mu = \mu_0$	$\mu > \mu_0$	Right-tailed
Is μ less than μ_0 ?	$\mu = \mu_0$	$\mu < \mu_0$	Left-tailed

The Six-Step Framework



Hypothesis Testing: Problem Setup

Question: Has the website traffic increased after marketing?

Given Data

- Historical average daily visits: $\mu = 10372$
- After marketing, a sample of 30 days is collected
- Sample mean: $\bar{x} = 10628$
- Sample standard deviation: $s = 402$

We want to determine whether the observed increase in daily visits is due to the marketing campaign or just random variation.

Hypothesis Testing: Hypotheses

Null Hypothesis

The mean daily visits have not changed.

$$H_0 : \mu = 10372$$

Any observed increase is attributed to random fluctuation.

Alternative Hypothesis

The mean daily visits have increased.

$$H_A : \mu > 10372$$

This is a **right-tailed hypothesis test**.

Hypothesis Testing: Test Statistic and Conclusion

Step 1: Assume H_0 is true.

Step 2: Compute the test statistic

$$z = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

$$z = \frac{10628 - 10372}{402/\sqrt{30}} \approx 3.49$$

Thus, the sample mean is about 3.49 standard errors above the hypothesized mean.

Step 3: Compute the p-value

$$p = P(Z \geq 3.49) \approx 0.0002$$

Conclusion

Since the p-value is very small, we reject H_0 .

Strong evidence that campaign increased the mean daily website visits.

Examples

1. When debating the State Appropriation for Penn State, the following question is asked: "Are the majority of students at Penn State from Pennsylvania?"

Variable: State (categorical) $\Rightarrow$ proportion p ; majority $> 50\%$: **right-tailed**

$$H_0 : p = 0.5 \quad \text{vs.} \quad H_a : p > 0.5$$

2. A consumer test agency wants to see whether the mean lifetime of a brand of tires is less than 42,000 miles. The tire manufacturer advertises that the average lifetime is at least 42,000 miles.

Variable: Lifetime (numerical) $\Rightarrow$ mean μ : **left-tailed**

$$H_0 : \mu = 42000 \quad \text{vs.} \quad H_a : \mu < 42000$$

3. The length of a certain lumber from a national home building store is supposed to be 8.5 feet. A builder wants to check whether the shipment of lumber she receives has a mean length different from 8.5 feet.

Variable: Length (numerical) $\Rightarrow$ mean μ : **two-tailed**

$$H_0 : \mu = 8.5 \quad \text{vs.} \quad H_a : \mu \neq 8.5$$

4. A political news company believes the national approval rating for the current president has fallen below 40

Variable: Approval (categorical) $\Rightarrow$ proportion p : **left-tailed**

$$H_0 : p = 0.4 \quad \text{vs.} \quad H_a : p < 0.4$$

Hypothesis Test for One-Sample Proportion

Key Condition:

If $np_0 > 5$ and $n(1 - p_0) > 5$, then $\hat{p}$ is approximately normal with:

$$\text{Mean} = p_0, \quad \text{SE} = \sqrt{\frac{p_0(1 - p_0)}{n}}$$

Null Hypothesis Assumption

We always assume $H_0: p = p_0$ is **true** when constructing the test statistic. The known value p_0 is used — *not* $\hat{p}$ — in the standard error.

Test Statistic

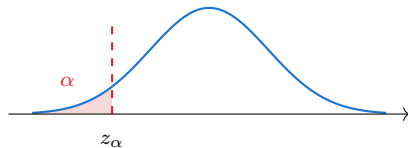
$$z^* = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1 - p_0)}{n}}}$$

Under H_0 , z^* follows a standard normal distribution $\mathcal{N}(0, 1)$.

Find p-value or rejection region

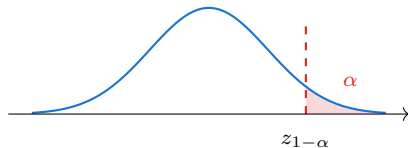
Rejection Region Approach:

Left-Tailed



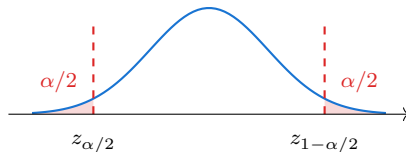
$H_a : p < p_0$
 Reject if $z^* \leq z_{\alpha}$

Right-Tailed



$H_a : p > p_0$
 Reject if $z^* \geq z_{1-\alpha}$

Two-Tailed



$H_a : p \neq p_0$
 Reject if $|z^*| \geq z_{1-\alpha/2}$

Find p-value or rejection region

P-Value Approach:

The probability of obtaining a test statistic **at least as extreme** as the observed one, *assuming H_0 is true*.

Test type	P-value formula
Left-tailed	$P(Z \leq z^*)$
Right-tailed	$P(Z \geq z^*)$
Two-tailed	$2 \times P(Z \geq z^*)$

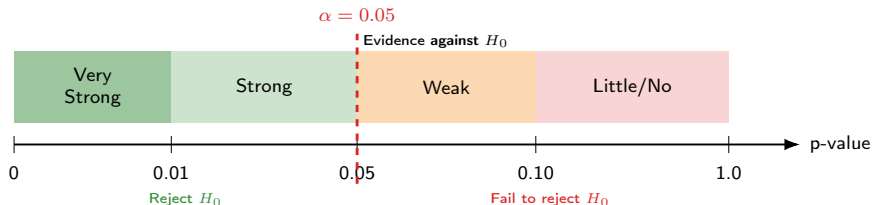
Decision Rule (same for all tests)

$$\text{p-value} \leq \alpha \Rightarrow \text{Reject } H_0$$

$$\text{p-value} > \alpha \Rightarrow \text{Fail to reject } H_0$$

Intuition: A small p-value means your data would be very unlikely if H_0 were true. Small enough (below α)? Then reject H_0 .

P-Value Strength Scale



Guideline

- p-value < 0.01 – Very strong evidence
- p-value < 0.05 – Strong evidence
- p-value < 0.10 – Weak evidence
- p-value ≥ 0.10 – Little / no evidence

Example 3

Problem:

A random sample of $n = 500$ Penn State students: 278 are from PA.
Can we conclude the proportion exceeds 0.5? ($\alpha = 0.05$)

Step 1 – Hypotheses & Conditions

$$H_0 : p = 0.5 \quad \text{vs} \quad H_a : p > 0.5 \quad (\text{right-tailed})$$

$$\text{Check: } np_0 = 500(0.5) = 250 \geq 5 \quad \checkmark \quad n(1 - p_0) = 250 \geq 5 \quad \checkmark$$

Step 2 – Significance Level $\alpha = 0.05$

Step 3 – Test Statistic

$$\hat{p} = \frac{278}{500} = 0.556, \quad z^* = \frac{0.556 - 0.5}{\sqrt{\frac{0.5 \times 0.5}{500}}} = \frac{0.056}{0.02236} = 2.504$$

***Intuition:** $\hat{p}$ is 2.5 standard errors above the hypothesised value of 0.5 – already sounds unlikely under H_0 !*

Example 3

Rejection Region Approach

Step 4: Right-tailed, $\alpha = 0.05$

Critical value: $z_{0.95} = 1.645$

Reject if $z^* \geq 1.645$

Step 5: $z^* = 2.504 > 1.645$

$\Rightarrow$ **Reject H_0**

Step 6:

At 5% significance, sufficient evidence that a **majority** of Penn State students are from PA.

P-Value Approach

Step 4:

$$\begin{aligned} \text{p-value} &= P(Z \geq 2.504) \\ &= 1 - 0.9938 = \mathbf{0.0062} \end{aligned}$$

Step 5: $0.0062 < 0.05 = \alpha$

$\Rightarrow$ **Reject H_0**

Step 6:

At 5% significance, sufficient evidence that a **majority** of Penn State students are from PA.

Both approaches give the same conclusion – always!

Try It!

Problem:

A company claims $\geq 60\%$ of graduate students buy online. A consumer group is sceptical. Sample of $n = 80$: only 22 have ever bought online. Test at $\alpha = 0.10$.

Your Turn – fill in each step:

- 1 Hypotheses? (Hint: left-tailed)
- 2 Check conditions ($np_0 \geq 5$?)
- 3 Compute z^* with $\hat{p} = 22/80 = 0.275$
- 4 Find rejection region / p-value
- 5 Decision?
- 6 Conclusion in context

Answer: $H_0 : p = 0.6$ vs $H_a : p < 0.6$; $z^* = -5.93$; p-value ≈ 0 ; **Reject** H_0 – evidence the true proportion is below 60%.

Test Statistic for a Mean

Conditions:

- Population is approximately **Normal**, or
- $n \geq 30$ (Central Limit Theorem applies)

One-Mean t -Statistic

$$t^* = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}} \sim t_{n-1}$$

Follows the t -distribution with $df = n - 1$ degrees of freedom.

Intuition: Same idea as the z -test, but we use s (estimated std. dev.) instead of σ , so we pay a small penalty via the t -distribution (heavier tails).

$\bar{x}$	sample mean
μ_0	hypothesised value
s	sample std. dev.
n	sample size

Example 4

Problem:

Standard lumber length = 8.5 ft. A builder measures $n = 61$ pieces:
 $\bar{x} = 8.3$ ft, $s = 1.2$ ft. Does the shipment differ from 8.5 ft? ($\alpha = 0.01$)

Step 1: Hypotheses & Conditions

$$H_0 : \mu = 8.5 \quad \text{vs} \quad H_a : \mu \neq 8.5 \quad (\text{two-tailed})$$

$$n = 61 \geq 30 \quad \checkmark$$

$$\text{Step 2: } \alpha = 0.01$$

Step 3:

$$t^* = \frac{8.3 - 8.5}{\frac{1.2}{\sqrt{61}}} = \frac{-0.2}{0.1537} = -1.30 \quad (df = 60)$$

Intuition: The sample mean is only 1.3 standard errors below 8.5 – not very far. We might not have enough evidence to reject H_0 .

Example 4

Rejection Region

Step 4: Two-tailed, $df = 60$

$$t_{\alpha/2, 60} = t_{0.005, 60} = 2.660$$

Reject if $|t^*| \geq 2.660$

Step 5: $|-1.30| = 1.30 < 2.660$

Fail to reject H_0

Step 6:

No significant evidence at 1% that mean length differs from 8.5 ft.

P-Value

Step 4:

$$p\text{-value} = 2P(T > 1.30)$$

From t -table ($df = 60$):

- $t = 1.296 \Rightarrow P = 0.10$

- $t = 1.671 \Rightarrow P = 0.05$

$$\Rightarrow 0.10 < p\text{-value} < 0.20$$

Step 5: $p\text{-value} > 0.01 = \alpha$

Fail to reject H_0

Step 6:

Same conclusion: insufficient evidence.

Try It!

Problem

A hospital claims the average weekend ER wait is 10 min. You sample $n = 40$ patients: $\bar{x} = 11$ min, $s = 3$ min.

Does the average *exceed* 10 min? ($\alpha = 0.05$)

Work through all six steps:

Step 1 $H_0 : \mu = 10$ vs $H_a : \mu > 10$; $n = 40 \geq 30$ ✓

Step 2 $\alpha = 0.05$

Step 3 $t^* = \frac{11 - 10}{3/\sqrt{40}} = \frac{1}{0.4743} = 2.11, df = 39$

Step 4 p-value = $P(T \geq 2.11) \approx 0.021$

Step 5 $0.021 < 0.05 \Rightarrow \text{Reject } H_0$

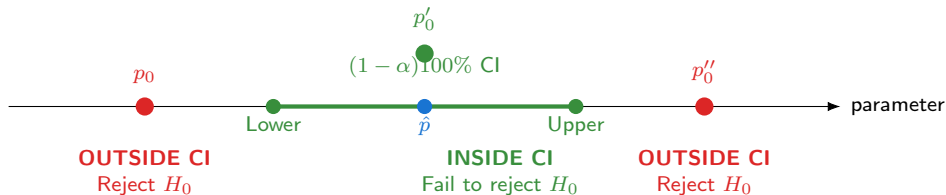
Step 6 Sufficient evidence ER wait exceeds 10 minutes.

Connecting CI to a Two-Tailed Test

Equivalence Theorem

For a two-tailed test at significance level α :

Reject $H_0 \iff$ hypothesised value p_0 (or μ_0) falls **outside** the $(1 - \alpha)100\%$ confidence interval.



Example 5

Problem Statement

Of 1500 sampled U.S. adults, 660 approved of the president's performance.

Test $H_0: p = 0.40$ vs. $H_a: p \neq 0.40$ at $\alpha = 0.05$.

Hypotheses, Conditions, and Test Statistic

Conditions: $np_0 = 600 > 5$ and $n(1 - p_0) = 900 > 5$ ✓ Normal approximation valid.

$$z^* = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1 - p_0)}{n}}} = \frac{0.44 - 0.40}{\sqrt{\frac{0.40(0.60)}{1500}}} = \mathbf{3.162}$$

P-value:

$$\text{p-value} = 2P(Z \geq 3.162) = 2(0.00078) = \mathbf{0.00156}$$

Decision and Conclusion

$\text{p-value} = 0.00156 < \alpha = 0.05 \Rightarrow$ **Reject H_0** . Sufficient evidence that the true approval proportion differs from 40%.

Intuition: The 95% CI from Lesson 5 is (0.415, 0.465). Since $p_0 = 0.40$ lies **outside** this interval, the CI and the two-sided test agree: reject H_0 . This connection holds for all two-sided tests and confidence intervals.

Common Mistakes to Avoid

#1 – Null hypothesis never has an inequality

Wrong: $H_0 : p \neq 0.5$, $H_a : p = 0.5$

Right: $H_0 : p = 0.5$, $H_a : p \neq 0.5$

#2 – Never “accept” the null

Wrong: “We accept H_0 ; the proportion equals 0.5.”

Right: “We fail to reject H_0 ; insufficient evidence against it.”

#3 – Always check conditions

Proportion: $np_0 \geq 5$ and $n(1 - p_0) \geq 5$

Mean: Normal population **or** $n \geq 30$

#4 – Match test direction to research question

“Exceeds” $\Rightarrow$ right-tailed; “Below” $\Rightarrow$ left-tailed; “Different from” $\Rightarrow$ two-tailed

R: One-Sample Proportion Test

R Code

```
# Using prop.test (normal approximation)
prop.test(x = 278, n = 500,
          p = 0.5,
          alternative = "greater",
          conf.level = 0.95,
          correct = FALSE)
```

Output includes: z-statistic, p-value, 95% CI

Expected Output (Example 3)

x	n	$\hat{p}$	95% Lower Bound	p-value
278	500	0.556	0.5195	0.006

Test Statistic for One Mean

Conditions (at least one must hold)

- Population is approximately **Normal**, or
- Sample size is **large** ($n > 30$)

One-Mean t -Test Statistic

$$t^* = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}} \sim t_{n-1} \text{ under } H_0$$

Degrees of freedom: $df = n - 1$

P-Value Formulas

Left-tailed: $P(t \leq t^*)$

Right-tailed: $P(t \geq t^*)$ where $t \sim t_{n-1}$

Two-tailed: $2P(t \geq |t^*|)$

Quick Reference – Two Tests Side-by-Side

	One-Proportion (z)	One-Mean (t)
Parameter	p	μ
Condition	$np_0 \geq 5; n(1 - p_0) \geq 5$	Normal pop. or $n \geq 30$
Statistic	$z^* = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1 - p_0)}{n}}}$	$t^* = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}}$
Distribution	$N(0, 1)$	t_{n-1}
Decision	$\text{p-value} \leq \alpha \Rightarrow \text{Reject } H_0$	

Example 6

Problem

Lumber is supposed to be 8.5 ft. A builder measures 61 pieces: $\bar{x} = 8.3$ ft, $s = 1.2$ ft. Test at $\alpha = 0.01$ whether the mean differs from 8.5 ft.

Solution

$H_0: \mu = 8.5$ vs. $H_a: \mu \neq 8.5$; $n = 61 > 30$ ✓

$$t^* = \frac{8.3 - 8.5}{1.2/\sqrt{61}} = -1.30, \quad df = 60$$

Rejection Region

$$t_{0.005, 60} = 2.660$$

Reject if $|t^*| \geq 2.660$

$$|-1.30| = 1.30 < 2.660$$

Fail to Reject H_0

P-Value

$$p\text{-value} = 2P(T > 1.30)$$

$$\approx 0.1 \text{ to } 0.2$$

$$> 0.01$$

Fail to Reject H_0

Conclusion

No significant evidence that mean lumber length differs from 8.5 ft at the 1% level.

Example 7

Problem

Hospital claims mean ER wait time is 10 min. Sample of 40 patients:
 $\bar{x} = 11$ min, $s = 3$ min. Test if wait time *exceeds* 10 min at $\alpha = 0.05$.

Solution

$H_0: \mu = 10$ vs. $H_a: \mu > 10$; $n = 40 > 30$ ✓

$$t^* = \frac{11 - 10}{3/\sqrt{40}} = \mathbf{2.11}, \quad df = 39$$

Rejection Region

$t_{0.05,39} = 1.685$

Reject if $t^* \geq 1.685$

$2.11 > 1.685$

Reject H_0

P-Value

$P(T \geq 2.11) \approx 0.021$

$0.021 < 0.05$

Reject H_0

Conclusion

Sufficient evidence at 5% that mean ER wait time **exceeds 10 minutes**.

R: One-Sample Mean t -Test

R Code

```
# Using t.test with summarized data
# For raw data: t.test(x, mu = 10, alternative = "greater")

# Simulating from summary stats
set.seed(42)
x <- c(rep(11, 40)) # placeholder; use actual data

t.test(x, mu = 10,
       alternative = "greater",
       conf.level = 0.95)
```

Expected Output (Example 7)

n	$\bar{x}$	s	t-value	p-value
40	11.000	3.000	2.11	0.021

Statistical vs. Practical Significance

Statistical Significance

Did the test reject H_0 ?

"Is the effect real?"

Influenced by:

sample size, variability, effect size

Practical Significance

Is the difference *meaningful*?

"Does it matter in practice?"

Influenced by:

context, cost, consequences

ER Wait Time – Same Example

Statistically significant $p\text{-value} = 0.021 < 0.05$

Practically significant? Is 1 extra minute worth a policy change?
Depends on hospital resources, patient impact, ...

Intuition: *With a very large n , even a tiny, unimportant difference can become statistically significant. Always report effect size alongside the*

Power, α , and β

Definitions

- α = P(Type I Error) — preset by researcher
- β = P(Type II Error) — depends on true effect
- **Power** = $1 - \beta$ = P(correctly rejecting false H_0)

Key Relationships

Action	Effect
Decrease α (fixed n)	β increases, Power decreases
Increase n (fixed α)	β decreases, Power increases

Intuition: The only way to decrease both α and β simultaneously is to **increase the sample size n** . Use R's `power.prop.test()` or `power.t.test()` to compute required sample sizes.

Try It

Problem

A CEO claims 80% of customers are “very satisfied”. A survey of 100 customers finds 75 satisfied. Test at $\alpha = 0.05$.

Solution

$H_0: p = 0.8$ vs. $H_a: p \neq 0.8$; conditions satisfied ✓

$$z^* = \frac{0.75 - 0.80}{\sqrt{0.8(0.2)/100}} = -1.25$$

$$\text{p-value} = 2P(Z > 1.25) = 2(0.1056) = 0.2112$$

$0.2112 > 0.05 \Rightarrow$ **Fail to Reject H_0**

Conclusion

Insufficient evidence at 5% to dispute the CEO's claim that 80% of customers are “very satisfied”.

Try It

Problem

A store claims checkout time ≤ 4 min. Six observations: 3.8, 5.3, 3.5, 4.5, 7.2, 5.1 min. Test at $\alpha = 0.10$.

Solution

$H_0: \mu = 4$ vs. $H_a: \mu > 4$; small $n = 6$, normality checked via Q-Q plot ✓

$$\bar{x} = 4.9, \quad s = 1.3282, \quad t^* = \frac{4.9 - 4}{1.3282/\sqrt{6}} = \mathbf{1.660}, \quad \text{df} = 5$$

$$\text{p-value} = P(T > 1.660) = 0.0789$$

$$0.0789 < 0.10 \Rightarrow \text{Reject } H_0$$

Conclusion

At 10% significance, sufficient evidence that mean checkout time **exceeds 4 minutes** — the store's claim is disputed.

Algorithm Selection Guide

Situation	Method
Categorical response, 1 group	One-proportion z -test
Quantitative response, 1 group	One-mean t -test
Small n , non-normal data	Non-parametric methods
Conditions not met ($np_0 < 5$)	Exact binomial test
Compare more than 2 groups	ANOVA (future lesson)

Key Concepts Summary

Logic & Setup

- Assume H_0 true; seek evidence against it
- H_0 has $=$; H_a has $<$, $>$, or $\neq$
- Test direction comes from research question

Errors

- Type I (α): reject true H_0
- Type II (β): miss false H_0
- Power $= 1 - \beta$; bigger n helps both

Two Approaches

- Rejection region: compare z^*/t^* to critical value
- P-value: compare p-value to α
- Both give identical conclusions

CI Connection

- Two-tailed only
- p_0/μ_0 outside CI $\Leftrightarrow$ reject H_0